CodeProctor

A Web-Based Coding Assessment and Malpractice Detection Platform

Database Schema

Team No: 15

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Initial Database Schema

This is the relational schema (PostgreSQL). It covers Authentication & RBAC, the Problem Bank, the Contest Engine (including per-contest roles and public/private visibility), the Judge, Proctoring, File Storage, Notifications, and Audit. Types are written in Postgres-style notation; primary keys use collision-resistant cuid strings (Prisma's default cuid()) except on the Better Auth tables, whose IDs are issued by the auth library.

Database Overview

PostgreSQL relational database with 31 tables covering authentication, the problem bank, the contest engine, the judge, proctoring, file storage, notifications, and audit.

#	Table	Group	Description
1	`user`	Auth	Registered users, authenticated via Better Auth
2	`session`	Auth	Active login sessions per user
3	`account`	Auth	Linked credential/OAuth accounts per user
4	`verification`	Auth	Short-lived email verification and reset tokens
5	`role`	Auth	Platform-level roles, e.g. admin, moderator
6	`user_role`	Auth	Many-to-many linking users to roles
7	`problem`	Problem Bank	Coding problems, public practice or contest-only
8	`tag`	Problem Bank	Topic tags for problems
9	`problem_tag`	Problem Bank	Many-to-many linking problems to tags
10	`test_case`	Problem Bank	Graded input/output pairs per problem
11	`problem_sample`	Problem Bank	Public example input/output shown to users
12	`language_template`	Problem Bank	Per-language starter code for a problem
13	`contest`	Contest Engine	Scheduled, proctored assessments
14	`contest_settings`	Contest Engine	Proctoring mode and access settings per contest
15	`contest_problem`	Contest Engine	Problems assigned to a contest, with points
16	`contest_member`	Contest Engine	Per-contest role: owner, moderator, or participant
17	`contest_invite`	Contest Engine	Email invites to a contest
18	`announcement`	Contest Engine	Messages broadcast to contest participants
19	`language`	Judge	Supported programming languages
20	`submission`	Judge	Submitted code and its verdict
21	`submission_test_case`	Judge	Per-test-case grading results
22	`execution_log`	Judge	Compile/run diagnostic logs per submission
23	`judge_job`	Judge	Worker/queue bookkeeping for judging
24	`proctor_session`	Proctoring	Proctoring session for a contest participant
25	`violation`	Proctoring	Detected proctoring violation
26	`snapshot`	Proctoring	Captured evidence frame for a violation

#	Table	Group	Description
27	face_metric	Proctoring	Periodic face/gaze detection reading
28	file	File Storage	Centralized storage for all uploaded/captured files
29	notification	Notifications	Notification events
30	notification_recipient	Notifications	Delivery/read state per notification recipient
31	audit_log	Audit	Log of administrative and moderation actions

1. user

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY	Unique identifier for every user; issued and managed by Better Auth.
name	TEXT	NOT NULL	Display name.
email	TEXT	NOT NULL, UNIQUE	Login identifier.
emailVerified	BOOLEAN	NOT NULL	Whether the user has completed email verification.
image	TEXT	NULL	Profile picture URL.
createdAt	TIMESTAMP	NOT NULL	Account creation time.
updatedAt	TIMESTAMP	NOT NULL	Last profile update.

2. session

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY	Unique identifier for the session.
expiresAt	TIMESTAMP	NOT NULL	When the session token expires.
token	TEXT	NOT NULL, UNIQUE	Session token presented by the client.
createdAt	TIMESTAMP	NOT NULL	Session creation time.
updatedAt	TIMESTAMP	NOT NULL	Last session refresh time.
ipAddress	TEXT	NULL	IP address the session was created from.
userAgent	TEXT	NULL	Client user agent string.
userId	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	Owning user.

3. account

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY	Unique identifier for the linked account.
accountId	TEXT	NOT NULL	Provider-side account identifier.
providerId	TEXT	NOT NULL	Authentication provider, e.g. "credential", "google".
userId	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	Owning user.

Column	Type	Constraints	Description
accessToken	TEXT	NULL	OAuth access token, if applicable.
refreshToken	TEXT	NULL	OAuth refresh token, if applicable.
idToken	TEXT	NULL	OAuth ID token, if applicable.
accessTokenExpiresAt	TIMESTAMP	NULL	Access token expiry.
refreshTokenExpiresAt	TIMESTAMP	NULL	Refresh token expiry.
scope	TEXT	NULL	Granted OAuth scopes.
password	TEXT	NULL	Hashed password, for credential-based accounts only.
createdAt	TIMESTAMP	NOT NULL	Link creation time.
updatedAt	TIMESTAMP	NOT NULL	Last update time.

4. verification

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY	Unique identifier for the verification record.
identifier	TEXT	NOT NULL	What is being verified, e.g. an email address.
value	TEXT	NOT NULL	The verification code or token value.
expiresAt	TIMESTAMP	NOT NULL	Expiry time of the token.
createdAt	TIMESTAMP	NULL	Creation time.
updatedAt	TIMESTAMP	NULL	Last update time.

5. role

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the role.
name	TEXT	NOT NULL, UNIQUE	Role name, e.g. "ADMIN", "USER", "MODERATOR".
description	TEXT	NULL	Optional human-readable description of the role.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Role creation time.

6. user_role

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the assignment row.
userId	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	The user being granted the role.
roleId	TEXT	FK -> role.id, NOT NULL, ON DELETE CASCADE	The role being granted.
assignedAt	TIMESTAMP	NOT NULL, DEFAULT now()	When the role was assigned.
UNIQUE	(userId, roleId)	constraint	A user can hold a given role only once.

7. problem

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the problem.
title	TEXT	NOT NULL	Problem title.
slug	TEXT	NOT NULL, UNIQUE	URL-friendly unique identifier.
description	TEXT	NOT NULL	Full problem statement (markdown).
difficulty	TEXT	NOT NULL, CHECK IN (EASY, MEDIUM, HARD)	Difficulty tag.
timeLimit	INTEGER	NOT NULL	Execution time limit, in milliseconds.
memoryLimit	INTEGER	NOT NULL	Memory limit, in kilobytes.
isPublic	BOOLEAN	NOT NULL, DEFAULT FALSE	TRUE = visible in the public practice bank.
isDeleted	BOOLEAN	NOT NULL, DEFAULT FALSE	Soft-delete flag; problems may still be referenced by past contests/submissions.
createdById	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	Author of the problem.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Creation time.
updatedAt	TIMESTAMP	NOT NULL	Last edit time (auto-updated).

8. tag

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the tag.
name	TEXT	NOT NULL, UNIQUE	Tag name.

9. problem_tag

Column	Type	Constraints	Description
problemId	TEXT	FK -> problem.id, ON DELETE CASCADE	Tagged problem.
tagId	TEXT	FK -> tag.id, ON DELETE CASCADE	Applied tag.
PRIMARY KEY	(problemId, tagId)	constraint	Composite primary key; a tag can be applied to a problem only once.

10. test_case

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for a test case.
problemId	TEXT	FK -> problem.id, NOT NULL, ON DELETE CASCADE	Owning problem.
input	TEXT	NOT NULL	Stdin fed to the submitted program.
expectedOutput	TEXT	NOT NULL	Expected stdout, compared against actual output.
isHidden	BOOLEAN	NOT NULL, DEFAULT FALSE	TRUE = not shown to participants, used only for grading.
weight	INTEGER	NOT NULL, DEFAULT 1	Enables partial scoring across test cases.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Creation time.

11. problem_sample

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the sample.
problemId	TEXT	FK -> problem.id, NOT NULL, ON DELETE CASCADE	Owning problem.
input	TEXT	NOT NULL	Sample input shown to users.
expectedOutput	TEXT	NOT NULL	Sample expected output shown to users.
explanation	TEXT	NULL	Optional walkthrough of the sample.
order	INTEGER	NOT NULL, DEFAULT 0	Display order among a problem's samples.

12. language_template

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the template.
problemId	TEXT	FK -> problem.id, NOT NULL, ON DELETE CASCADE	Owning problem.
languageId	TEXT	FK -> language.id, NOT NULL, ON DELETE CASCADE	Language the template is for.
boilerplate	TEXT	NOT NULL	Starter code shown in the editor.
UNIQUE	(problemId, languageId)	constraint	At most one template per problem/language pair.

13. contest

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the contest.
title	TEXT	NOT NULL	Contest title.
description	TEXT	NULL	Optional contest description / rules.
startTime	TIMESTAMP	NOT NULL	Scheduled start.
endTime	TIMESTAMP	NOT NULL	Scheduled end.
duration	INTEGER	NOT NULL	Contest duration, in minutes.

Column	Type	Constraints	Description
isDeleted	BOOLEAN	NOT NULL, DEFAULT FALSE	Soft-delete flag.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Creation time.
updatedAt	TIMESTAMP	NOT NULL	Last edit time (auto-updated).

14. contest_settings

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the settings row.
contestId	TEXT	FK -> contest.id, UNIQUE, NOT NULL, ON DELETE CASCADE	Owning contest (one-to-one).
proctoringMode	TEXT	NOT NULL, DEFAULT 'NORMAL', CHECK IN (NORMAL, FULL_SCREEN, CV_PROCTORED)	Proctoring level selected for the contest.
requireCamera	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether camera access is required to participate.
requireFullScreen	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether full-screen mode is enforced.
detectTabSwitch	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether tab-switch detection is enabled.
detectCopyPaste	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether copy/paste detection is enabled.
password	TEXT	NULL	Optional join password.
allowLateJoin	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether participants may join after start time.
isPublic	BOOLEAN	NOT NULL, DEFAULT TRUE	Public = listed / discoverable; FALSE = invite-only.

15. contest_problem

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the link row.
contestId	TEXT	FK -> contest.id, NOT NULL, ON DELETE CASCADE	Owning contest.
problemId	TEXT	FK -> problem.id, NOT NULL, ON DELETE CASCADE	Assigned problem.
order	INTEGER	NOT NULL, DEFAULT 0	Display order within the contest.
points	INTEGER	NOT NULL, DEFAULT 100	Points awarded for this problem in this contest.
UNIQUE	(contestId, problemId)	constraint	A problem can be added to a contest only once.

16. contest_member

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the membership row.
contestId	TEXT	FK -> contest.id, NOT NULL, ON DELETE CASCADE	Contest joined.
userId	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	Member.
role	TEXT	NOT NULL, DEFAULT 'PARTICIPANT', CHECK IN (OWNER, MODERATOR, PARTICIPANT)	Per-contest role. OWNER = created the contest, full control. MODERATOR = manages problems/announcements. PARTICIPANT = attends and accumulates score.
invitedById	TEXT	FK -> user.id, NULL, ON DELETE SET NULL	Who added this member, for MODERATOR / invited PARTICIPANT rows.
joinedAt	TIMESTAMP	NOT NULL, DEFAULT now()	Registration time.

Column	Type	Constraints	Description
bestScore	INTEGER	NOT NULL, DEFAULT 0	Best score achieved, used for leaderboard ranking.
penalty	INTEGER	NOT NULL, DEFAULT 0	Accumulated penalty, used as a tiebreaker.
lastAcceptedSubmission	TIMESTAMP	NULL	Timestamp of the last accepted submission.
isDisqualified	BOOLEAN	NOT NULL, DEFAULT FALSE	Set TRUE once disqualified.
disqualifiedReason	TEXT	NULL	Free-text reason, typically referencing the violation timeline.
disqualifiedAt	TIMESTAMP	NULL	When the disqualification was recorded.
UNIQUE	(contestId, userId)	constraint	A user can hold exactly one role in a given contest.

17. contest_invite

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the invite.
contestId	TEXT	FK -> contest.id, NOT NULL, ON DELETE CASCADE	Contest the invite grants access to.
email	TEXT	NOT NULL	Invited email address (may not yet correspond to a registered user).
role	TEXT	NOT NULL, DEFAULT 'PARTICIPANT', CHECK IN (OWNER, MODERATOR, PARTICIPANT)	Role the invitee will receive on acceptance; owners can invite directly as MODERATOR.
status	TEXT	NOT NULL, DEFAULT 'PENDING', CHECK IN (PENDING, ACCEPTED, DECLINED, EXPIRED)	Invite lifecycle state.
invitedAt	TIMESTAMP	NOT NULL, DEFAULT now()	Invite creation time.
respondedAt	TIMESTAMP	NULL	When the invitee responded.

Column	Type	Constraints	Description
UNIQUE	(contestId, email)	constraint	An email can be invited to a given contest only once.

18. announcement

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the announcement.
contestId	TEXT	FK -> contest.id, NOT NULL, ON DELETE CASCADE	Contest the announcement belongs to.
message	TEXT	NOT NULL	Announcement body.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Creation time.

19. language

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the language.
name	TEXT	NOT NULL, UNIQUE	Display name, e.g. "Python 3.11".
slug	TEXT	NOT NULL, UNIQUE	URL/API-friendly identifier, e.g. "python3".
dockerImage	TEXT	NOT NULL	Docker image used to compile/run submissions in this language.
isActive	BOOLEAN	NOT NULL, DEFAULT TRUE	Whether the language is currently offered.

20. submission

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the submission.
userId	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	Submitting user.
problemId	TEXT	FK -> problem.id, NOT NULL, ON DELETE CASCADE	Problem being solved.
contestId	TEXT	FK -> contest.id, NULL, ON DELETE CASCADE	NULL = public practice submission; set = graded contest submission.

Column	Type	Constraints	Description
languageId	TEXT	FK -> language.id, NOT NULL	Submission language.
sourceCode	TEXT	NOT NULL	Submitted source code.
status	TEXT	NOT NULL, DEFAULT 'QUEUED', CHECK IN (QUEUED, RUNNING, COMPLETED)	Judging lifecycle state.
verdict	TEXT	NULL, CHECK IN (ACCEPTED, WRONG_ANSWER, TIME_LIMIT_EXCEEDED, MEMORY_LIMIT_EXCEEDED, RUNTIME_ERROR, COMPILATION_ERROR)	Aggregate/final verdict.
executionTime	INTEGER	NULL	Worst-case runtime across test cases, in milliseconds.
memoryUsed	INTEGER	NULL	Worst-case memory used across test cases, in kilobytes.
submittedAt	TIMESTAMP	NOT NULL, DEFAULT now()	Submission time; used as the leaderboard tiebreaker.
judgedAt	TIMESTAMP	NULL	When judging completed.

21. submission_test_case

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the result row.
submissionId	TEXT	FK -> submission.id, NOT NULL, ON DELETE CASCADE	Owning submission.
testCaseId	TEXT	FK -> test_case.id, NOT NULL, ON DELETE CASCADE	Test case graded.
verdict	TEXT	NOT NULL, CHECK IN (ACCEPTED, WRONG_ANSWER, TIME_LIMIT_EXCEEDED, MEMORY_LIMIT_EXCEEDED, RUNTIME_ERROR, COMPILATION_ERROR)	Result for this specific test case.
executionTime	INTEGER	NULL	Runtime for this test case, in milliseconds.
memoryUsed	INTEGER	NULL	Memory used for this test case, in kilobytes.
actualOutput	TEXT	NULL	Actual stdout produced, for debugging/display.

Column	Type	Constraints	Description
UNIQUE	(submissionId, testCaseId)	constraint	One result per test case per submission.

22. execution_log

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the log entry.
submissionId	TEXT	FK -> submission.id, NOT NULL, ON DELETE CASCADE	Owning submission.
stage	TEXT	NOT NULL	Pipeline stage: "queue" | "compile" | "run".
message	TEXT	NOT NULL	Log message.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Log timestamp.

23. judge_job

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the job.
submissionId	TEXT	FK -> submission.id, UNIQUE, NOT NULL, ON DELETE CASCADE	Submission this job judges (one-to-one).
queuedAt	TIMESTAMP	NOT NULL, DEFAULT now()	When the job was queued.
startedAt	TIMESTAMP	NULL	When a worker picked up the job.
completedAt	TIMESTAMP	NULL	When the job finished.
workerId	TEXT	NULL	Identifier of the worker that ran the job.
attempts	INTEGER	NOT NULL, DEFAULT 0	Number of execution attempts.

24. proctor_session

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the session.
participantId	TEXT	FK -> contest_member.id, UNIQUE, NOT NULL, ON DELETE CASCADE	The participating contest_member (one-to-one).
startedAt	TIMESTAMP	NOT NULL, DEFAULT now()	Session start time.
endedAt	TIMESTAMP	NULL	Session end time.
cameraGranted	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether camera permission was granted.
micGranted	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether microphone permission was granted.

25. violation

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the violation.
proctorSessionId	TEXT	FK -> proctor_session.id, NOT NULL, ON DELETE CASCADE	Session in which the violation occurred.
type	TEXT	NOT NULL, CHECK IN (TAB_SWITCH, FULLSCREEN_EXIT, NO_FACE, MULTIPLE_FACES, LOOKING_AWAY, PASTE)	Kind of violation detected.
severity	TEXT	NOT NULL, CHECK IN (LOW, MEDIUM, HIGH)	Used to visually prioritize entries in the timeline.
timestamp	TIMESTAMP	NOT NULL, DEFAULT now()	When the violation occurred.
metadata	JSONB	NULL	Additional structured detail about the violation.

26. snapshot

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the snapshot.
violationId	TEXT	FK -> violation.id, NOT NULL, ON DELETE CASCADE	Violation this snapshot is evidence for.
fileId	TEXT	FK -> file.id, NOT NULL	Stored image file (no cascade).
capturedAt	TIMESTAMP	NOT NULL, DEFAULT now()	Capture time.

27. face_metric

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the reading.
proctorSessionId	TEXT	FK -> proctor_session.id, NOT NULL, ON DELETE CASCADE	Session the reading belongs to.
timestamp	TIMESTAMP	NOT NULL, DEFAULT now()	When the reading was taken.
faceCount	INTEGER	NOT NULL	Number of faces detected in frame.
gazeDeviationDeg	DOUBLE PRECISION	NULL	Measured gaze deviation, in degrees.

28. file

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the file.
bucket	TEXT	NOT NULL	Storage bucket the object lives in.
key	TEXT	NOT NULL	Object key/path within the bucket.
url	TEXT	NOT NULL	Publicly resolvable (or signed) URL.
uploadedById	TEXT	FK -> user.id, NULL, ON DELETE SET NULL	User who uploaded the file, if applicable.
mimeType	TEXT	NOT NULL	MIME type of the file.

Column	Type	Constraints	Description
size	INTEGER	NOT NULL	File size, in bytes.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Upload time.

29. notification

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the notification.
type	TEXT	NOT NULL	Notification kind, e.g. "CONTEST_STARTED", "INVITATION", "DISQUALIFICATION", "SUBMISSION_JUDGED".
title	TEXT	NOT NULL	Notification title.
message	TEXT	NOT NULL	Notification body.
metadata	JSONB	NULL	Additional structured payload.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	Creation time.

30. notification_recipient

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the recipient row.
notificationId	TEXT	FK -> notification.id, NOT NULL, ON DELETE CASCADE	Notification being delivered.
userId	TEXT	FK -> user.id, NOT NULL, ON DELETE CASCADE	Recipient.
isRead	BOOLEAN	NOT NULL, DEFAULT FALSE	Whether the recipient has read it.
readAt	TIMESTAMP	NULL	When it was read.
UNIQUE	(notificationId, userId)	constraint	A notification is delivered to a given user only once.

31. audit_log

Column	Type	Constraints	Description
id	TEXT / PK	PRIMARY KEY, DEFAULT cuid()	Unique identifier for the log entry.
actorId	TEXT	FK -> user.id, NULL, ON DELETE SET NULL	User who performed the action, if known.
action	TEXT	NOT NULL	Action taken, e.g. "DELETE_PROBLEM", "UPDATE_CONTEST", "BAN_USER", "CHANGE_ROLE".
entityType	TEXT	NOT NULL	Type of entity affected.
entityId	TEXT	NOT NULL	ID of the entity affected.
before	JSONB	NULL	Entity state before the action.
after	JSONB	NULL	Entity state after the action.
createdAt	TIMESTAMP	NOT NULL, DEFAULT now()	When the action occurred.

Entity Relationships

- user (1) — (N) problem, via problem.createdById
- user (1) — (N) contest_member (N) — (1) contest — a many-to-many between users and contests, carrying a per-contest role (OWNER / MODERATOR / PARTICIPANT) instead of a fixed ownerId column
- role (1) — (N) user_role (N) — (1) user — a many-to-many between users and platform roles
- problem (1) — (N) test_case, via test_case.problemId
- problem (1) — (N) problem_sample, via problem_sample.problemId
- problem (1) — (N) contest_problem (N) — (1) contest — a many-to-many between problems and contests, carrying order and points
- problem (1) — (N) problem_tag (N) — (1) tag — a many-to-many between problems and tags
- contest (1) — (1) contest_settings, via contest_settings.contestId
- contest (1) — (N) contest_invite, via contest_invite.contestId
- contest (1) — (N) announcement, via announcement.contestId
- user (1) — (N) submission, via submission.userId; submission.contestId is NULL for public practice, set for graded attempts
- submission (1) — (N) submission_test_case (N) — (1) test_case — the full per-test-case breakdown
- submission (1) — (0..1) judge_job, via judge_job.submissionId
- contest_member (1) — (0..1) proctor_session, via proctor_session.participantId (PARTICIPANT rows only)
- proctor_session (1) — (N) violation, via violation.proctorSessionId
- proctor_session (1) — (N) face_metric, via face_metric.proctorSessionId
- violation (1) — (N) snapshot (N) — (1) file — evidence frames for a violation

- user (1) — (N) file, via file.uploadedById
- notification (1) — (N) notification_recipient (N) — (1) user — a many-to-many between notifications and users, carrying read state
- user (1) — (N) audit_log, via audit_log.actorId